

Nguyen Thanh Long

Full-Stack Engineer · LLM & agent systems

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SUMMARY

Engineer building on LLM APIs in TypeScript: an MCP gateway exposing a production IoT platform to Claude, ChatGPT and n8n as fifteen authorised tools, and a writing system whose token cost stays bounded however long the document grows. My day job is the cloud side of that same platform, a few million API requests a day.

EXPERIENCE

Full-Stack Engineer at Rogo Solutions JSC

Jan 2024 to present · Hanoi

Own the cloud side of a multi-tenant IoT platform end to end: the partner dashboard, the API, the message pipeline, and the deployments. It runs at a few million API requests a day, with around 15,000 concurrent MQTT connections and roughly 12 million device messages a day.

- Build the partner dashboard in **Next.js** and TypeScript, with **React Query** for server state, Ant Design and Tailwind for the interface, and internationalisation. A single deployment serves multiple partner projects, configurable per project.
- Attribute-based authorisation** in which the permission set is encoded into a compact **custom claim on the JWT**, so a request is authorised from the token itself instead of a database lookup on every call. The React app derives the same rules through **CASL**, so the interface and the API agree on what is allowed.
- Device state streaming on **Kafka** with **MQTT** ingestion: a consumer that coalesces repeated updates into batched writes and commits offsets after the write, autoscaled by **KEDA** on consumer group lag.
- Run the platform on **Kubernetes** and **AWS ECS**, with **Prometheus**, **Loki** and **Grafana** for metrics and logs, and the staging to production path on GitHub Actions, GitLab CI and **ArgoCD**.

SELECTED PROJECTS

IoT Cloud MCP (TypeScript, NestJS, MCP, Redis) · github.com/dadadadas111/iot-cloud-mcp
An MCP gateway that lets Claude, ChatGPT and n8n query and control a production IoT platform: fifteen tools over JSON-RPC, with the device REST API behind them. Multi-tenant, and sessions in Redis are scoped per project and per user — a tool call is authorised by the session, not trusted because the model asked for it.

Novel Studio (TypeScript, NestJS, Anthropic SDK, PostgreSQL with pgvector) · github.com/dadadadas111/novel-studio
Token cost normally grows with the story until it stops being affordable. Three tiers keep the prompt bounded at any length: a canon store of extracted facts, a rolling summary that is recursively recompressed, and top-k pgvector retrieval. The model never sees the raw novel.

MathSketch Tutor (TypeScript, NestJS, mathjs, MCP) · github.com/dadadadas111/mathsketch-tutor
Feedback on a handwritten maths problem. A computer-algebra engine, exposed to the model as an MCP tool, decides whether a step is right and the model only writes the explanation — so it cannot invent a correction that is wrong.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, C#, Java, SQL
Frontend	React, Next.js, Tailwind CSS, Ant Design, React Query, React Native (Expo)
AI & agents	Anthropic Claude SDK, MCP (servers and tools), agentic tool use, context engineering, pgvector RAG, n8n
Backend	Node.js, NestJS, Express, .NET (ASP.NET Core), REST APIs, WebSockets, Microservices
Data	PostgreSQL, MySQL, SQL Server, MongoDB, Redis, pgvector, Supabase
Infrastructure	Docker, Kubernetes, Linux, Nginx, AWS (ECS, S3), CI/CD (GitHub Actions, GitLab CI), Git
Messaging & auth	Kafka, MQTT, Firebase Auth, Keycloak, JWT

EDUCATION & ADDITIONAL

B.Sc. in Software Engineering, FPT University

Graduated 2026

Certifications: Project Management Principles (UC Irvine, 2025) · Software Development Lifecycle Specialization (University of Minnesota, 2024).
Coach competitive-programming students for national Informatics examinations, weekly since 2024. Freelance backend work alongside full-time employment. English: professional working proficiency.